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Prints:

Revisão:

- Parâmetros:

$$\text{real } Y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki} + \mu_i \quad \left. \begin{array}{l} \mu_i = Y_i - \hat{Y}_i \\ \text{fitte } \hat{Y}_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_K X_{Ki} \end{array} \right\} \begin{array}{l} \text{MQO:} \\ \sum_{i=1}^n \mu_i = 0 \quad \text{I} \\ \sum_{i=1}^n \mu_i^2 = \min \quad \text{II} \end{array}$$

$$\sum_{i=1}^n \mu_i^2 = \min \quad \text{II}$$

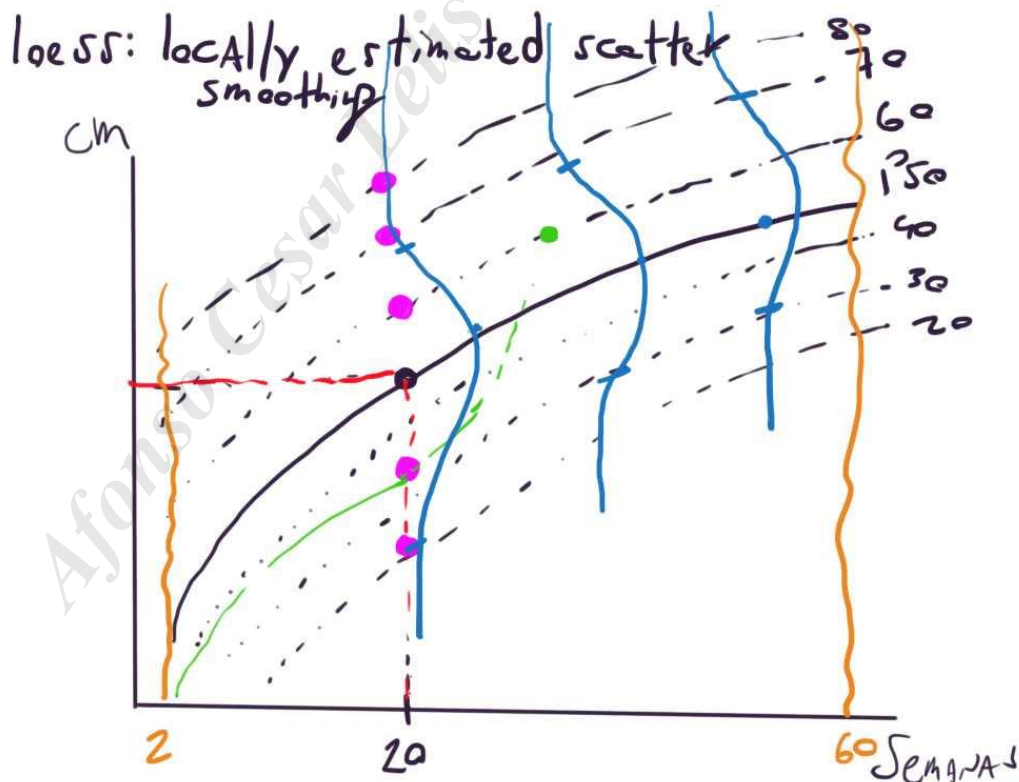
- R^2

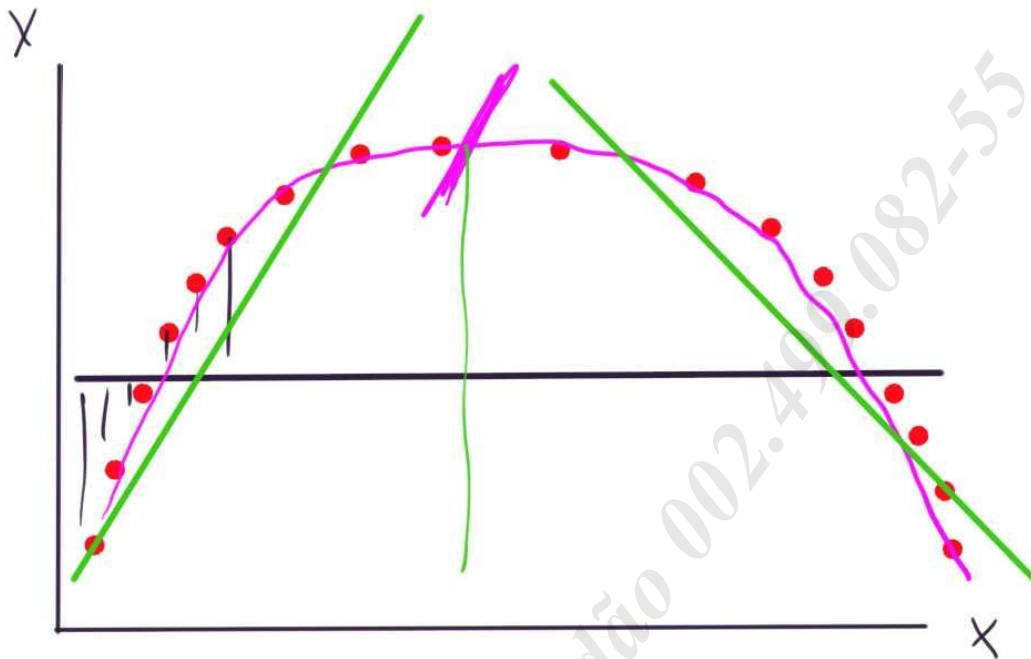
- Teste F.

- testes t. $\rightarrow \alpha$ fica!

- ICs p/ fins construção cenários preditivos.

- Dummies (n-1)





Modelo Linear:

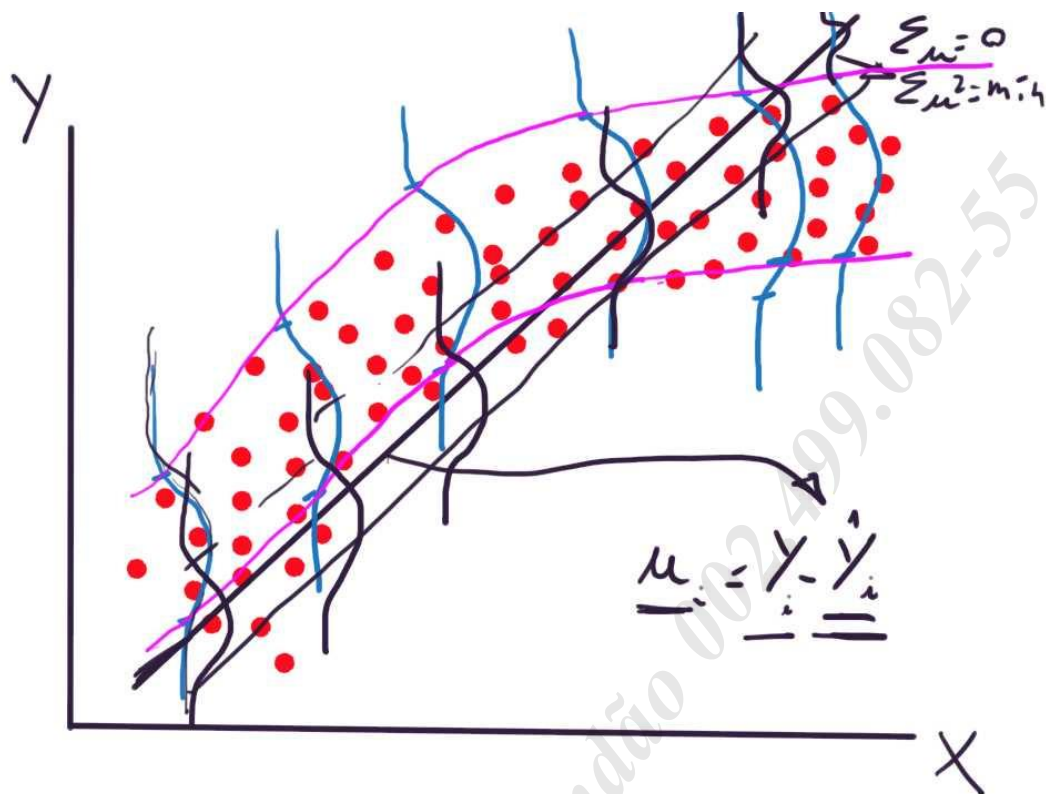
$$\hat{Comp}_i = 43,10 + 0,94 \times idade_i$$

$$R^2 = 90,27\%$$

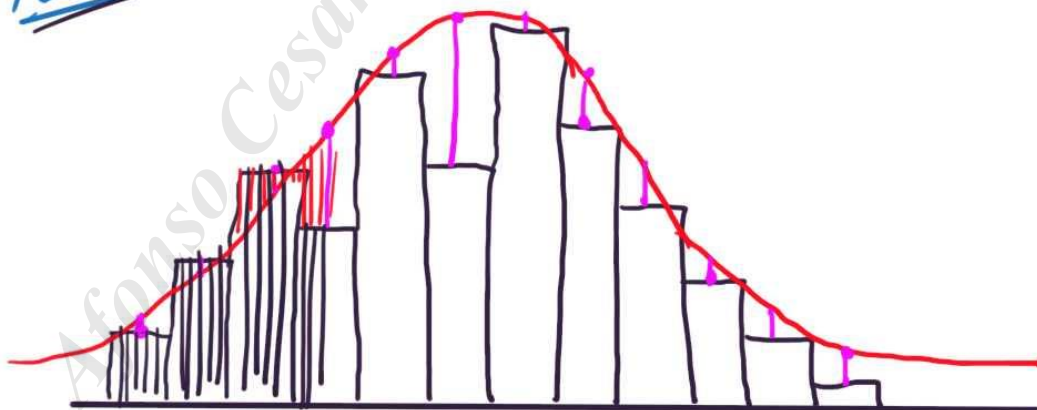
Modelo Não Linear: (Box-Cox):

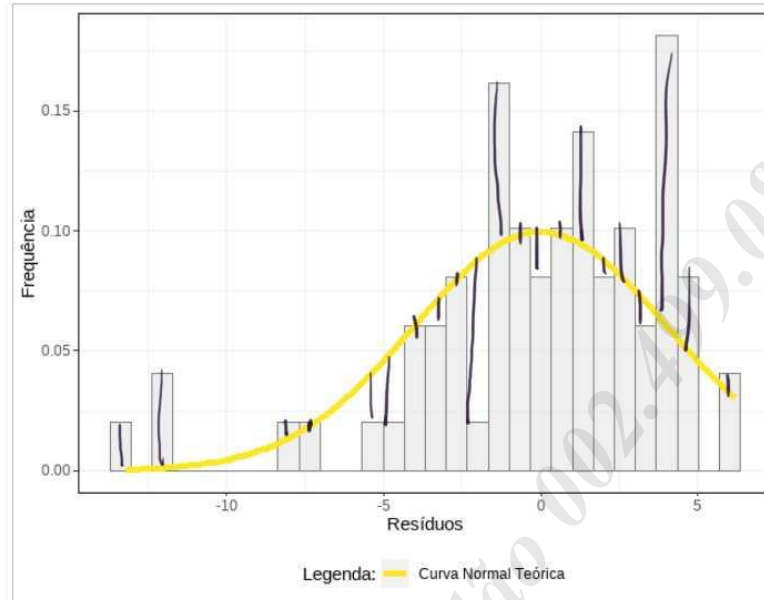
$$\frac{\hat{Comp}_i^{2,659} - 1}{2,659} = 4995,16 + 947,23 \times idade_i$$

$$R^2 = 96,20\%$$



Normal: ($p\text{-value} > 0,05$) $H_0: \neq s$ não est. sig. (no)
Não Normal: ($p\text{-value} < 0,05$) $H_1: \neq s$ est. sig.





Box-Cox (1964):

$$Y^* = \frac{Y^{\lambda} - 1}{\lambda}$$

λ : parâmetro Box-Cox.

$$-\infty \leq \lambda \leq \infty$$

$\lambda = 1$: linear

$\lambda = 2$: quadrática

$\lambda = 3$: cúbica

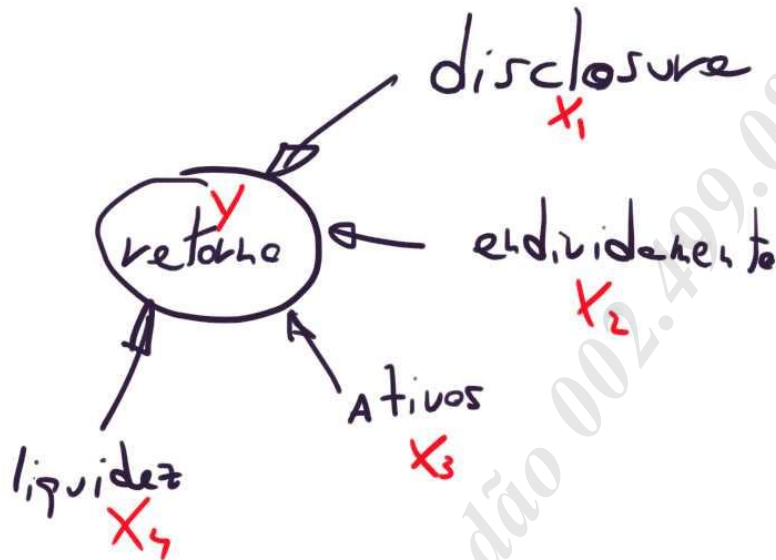
$\lambda = -1$: inversa

$\lambda = 0,5$: raiz

$\lambda = -0,5$: raiz inv.

$\lambda = 0$: $\ln(Y)$ (ex. expansão de Taylor).

Computat Global (Wharton/Penn):



$$\frac{\hat{\text{retorno}}_i - (-0,022)}{-0,022} = 2,82 + 0,00004 \frac{\text{Ativos}_i}{4000} + 0,939 \frac{\text{liquidez}_i}{14} + 0,0031 \frac{\text{disclosure}_i}{50}$$

3,7020

$$R^2_{aj} = 1 - \frac{n-1}{n-1-k} (1-R^2).$$

n : Amostra.

k : variáveis explicativas.

$$R^2_y = 1 - \frac{124-1}{124-1-3} (1-0.9799)$$

SUGESTÃO DE LEITURA:

- Box, G.E.P.; Cox, D.R. 1964. **An Analysis of Transformations**. Journal of the Royal Statistical Society.